

Resolving Deep Nodes in an Ancient Radiation of Neotropical Fishes in the Presence of Conflicting Signals from Incomplete Lineage Sorting

SUPPLEMENTARY MATERIAL

Table S1. Concordance factors and their 95% CI for the most frequent bipartitions in the concordance tree inferred from the Bayesian concordance analysis in BUCKy with values of $\alpha=1$, 5, 10 and ∞ .

Bipartition	$\alpha=1$	$\alpha=5$	$\alpha=10$	$\alpha=\infty$
Gymnotiformes ...	0.961 (0.954-0.970)	0.961 (0.951-0.970)	0.96 (0.951-0.967)	0.848 (0.826-0.872)
Apteronotidae ...	0.981 (0.973-0.989)	0.979 (0.970-0.986)	0.981 (0.973-0.989)	0.937 (0.918-0.954)
Sternopygidae ...	0.558 (0.527-0.601)	0.565 (0.541-0.590)	0.571 (0.541-0.598)	0.347 (0.315-0.380)
Pulseoidea ...	0.386 (0.353-0.435)	0.402 (0.372-0.438)	0.398 (0.353-0.440)	0.34 (0.304-0.375)
Gymnotidae ...	0.29 (0.242-0.342)	0.277 (0.245-0.312)	0.285 (0.236-0.326)	0.157 (0.128-0.188)
Rhamphichthyoidea ...	0.908 (0.886-0.924)	0.903 (0.872-0.924)	0.908 (0.886-0.924)	0.719 (0.690-0.747)
Pulseoidea ...	0.386 (0.353-0.435)	0.402 (0.372-0.438)	0.398 (0.353-0.440)	0.34 (0.304-0.375)

Table S2. Bootstrap support values recovered for the major nodes of the Gymnotiformes species tree inferred in ASTRAL-II for each one of the filtered and non-filtered datasets. Asterisks (*) indicates incongruence among datasets.

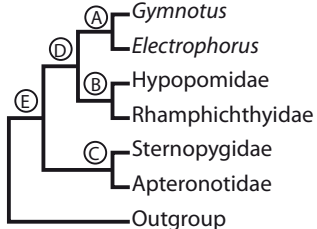
			Node							
Dataset			A	B	C	D	E	%RF	Δ%RF	
Full taxon set	Non-Filtered	All loci	60	100	52	100	100	86.10%	0.00%	
	75% complete	1 st + 2 nd quartiles	Parsimony Informative	70	100	65	100	100	82.76%	3.34%
			Molecular Clock	*	100	40	100	100	87.34%	-1.23%
			Phylogenetic Informativeness	64	100	61	100	100	82.87%	3.23%
			QIRP	62	100	73	100	100	83.36%	2.74%
		1 st quartile	Parsimony Informative	56	100	79	100	100	80.43%	5.67%
		Molecular Clock	*	100	69	100	100	88.06%	-1.96%	
		Phylogenetic Informativeness	58	100	66	100	100	80.72%	5.38%	
		QIRP	61	100	88	100	100	80.70%	5.40%	
	Reduced taxon set	Non-Filtered	All loci	55	100	58	100	100	74.15%	0.00%
	75% complete	1 st + 2 nd quartiles	Parsimony Informative	43	100	73	100	100	67.91%	6.23%
			Molecular Clock	*	100	88	100	100	77.59%	-3.45%
			Phylogenetic Informativeness	45	100	64	100	100	68.43%	5.72%
QIRP			47	100	49	100	100	67.62%	6.53%	
1 st quartile		Parsimony Informative	48	100	28	100	100	61.96%	12.18%	
		Molecular Clock	56	100	59	97	100	79.52%	-5.37%	
		Phylogenetic Informativeness	68	100	55	100	100	63.12%	11.03%	
		QIRP	41	100	63	100	100	61.65%	12.49%	
Reduced taxon set		Non-Filtered	All loci	75	100	69	100	100	55.78%	0.00%
100% complete		1 st + 2 nd quartiles	Parsimony Informative	71	100	73	100	100	51.39%	4.39%
			Molecular Clock	*	100	59	99	100	58.15%	-2.37%
			Phylogenetic Informativeness	53	100	69	100	100	51.16%	4.61%
	QIRP		67	100	72	100	100	50.12%	5.66%	
	1 st quartile	Parsimony Informative	*	100	70	100	100	50.91%	4.87%	
		Molecular Clock	*	100	61	80	100	63.21%	-7.43%	
		Phylogenetic Informativeness	59	100	83	100	100	50.63%	5.15%	
		QIRP	70	100	57	100	100	50.29%	5.49%	

Table S3. Base frequencies and percentage of GC (%GC) content in each major lineage and family across all filtered and unfiltered data sets of the full taxon set. Chi-Square homogeneity tests *P*-values in bold indicate significance at *P*<0.05.

	Base								
Taxon	A	C	G	T	GC	No. sites	χ^2	d.f.	P
Non-Filtered									
All	0.26	0.24	0.24	0.26	0.48	300099.79	1851.36	138	0.000
Ingroup	0.26	0.24	0.24	0.26	0.48	302792.26	770.71	123	0.000
Apteronotidae	0.26	0.24	0.24	0.26	0.48	313552.13	49.31	42	0.204
Gymnotidae	0.26	0.24	0.24	0.26	0.48	300842.38	55.11	21	0.000
<i>Gymnotus</i>	0.26	0.24	0.24	0.26	0.47	287067.67	40.85	15	0.000
Hypopomidae	0.26	0.24	0.24	0.26	0.48	304828.00	5.56	9	0.783
Rhamphichthyidae	0.26	0.24	0.24	0.26	0.48	289855.33	164.02	6	0.000
Rhamphichthyoidea	0.26	0.24	0.24	0.26	0.48	298411.14	184.93	18	0.000
Sternopygidae	0.26	0.24	0.24	0.26	0.48	293198.00	164.78	33	0.000
Parsimony Informative 1 st + 2 nd quartiles									
All	0.27	0.23	0.23	0.27	0.46	183659.15	1313.43	138	0.000
Ingroup	0.27	0.23	0.23	0.27	0.47	185336.93	634.22	123	0.000
Apteronotidae	0.27	0.23	0.23	0.27	0.46	191692.67	36.32	42	0.718
Gymnotidae	0.27	0.23	0.23	0.27	0.46	183499.75	26.11	21	0.202
<i>Gymnotus</i>	0.27	0.23	0.23	0.27	0.46	174743.17	18.63	15	0.231
Hypopomidae	0.26	0.24	0.24	0.26	0.47	185717.75	9.50	9	0.393
Rhamphichthyidae	0.26	0.24	0.23	0.27	0.47	175875.67	135.53	6	0.000
Rhamphichthyoidea	0.26	0.24	0.24	0.26	0.47	181499.71	158.92	18	0.000
Sternopygidae	0.27	0.23	0.23	0.27	0.47	180855.42	110.56	33	0.000
Molecular Clock 1 st + 2 nd quartiles									
All	0.27	0.23	0.23	0.27	0.46	183659.15	1313.43	138	0.000
Ingroup	0.27	0.23	0.23	0.27	0.47	185336.93	634.22	123	0.000
Apteronotidae	0.27	0.23	0.23	0.27	0.46	191692.67	36.32	42	0.718
Gymnotidae	0.27	0.23	0.23	0.27	0.46	183499.75	26.11	21	0.202
<i>Gymnotus</i>	0.27	0.23	0.23	0.27	0.46	174743.17	18.63	15	0.231
Hypopomidae	0.26	0.24	0.24	0.26	0.47	185717.75	9.50	9	0.393
Rhamphichthyidae	0.26	0.24	0.23	0.27	0.47	175875.67	135.53	6	0.000
Rhamphichthyoidea	0.26	0.24	0.24	0.26	0.47	181499.71	158.92	18	0.000
Sternopygidae	0.27	0.23	0.23	0.27	0.47	180855.42	110.56	33	0.000
Phylogenetic Informativeness 1 st + 2 nd quartiles									
All	0.27	0.23	0.23	0.27	0.46	183222.21	1304.63	138	0.000
Ingroup	0.27	0.23	0.23	0.27	0.47	185048.64	636.42	123	0.000
Apteronotidae	0.27	0.23	0.23	0.27	0.46	191334.93	41.81	42	0.479
Gymnotidae	0.27	0.23	0.23	0.27	0.46	183600.88	25.98	21	0.207
<i>Gymnotus</i>	0.27	0.23	0.23	0.27	0.46	175443.17	16.49	15	0.350
Hypopomidae	0.26	0.24	0.24	0.26	0.47	185589.75	8.14	9	0.520
Rhamphichthyidae	0.26	0.24	0.23	0.27	0.47	175619.67	142.11	6	0.000
Rhamphichthoidea	0.26	0.24	0.23	0.26	0.47	181316.86	161.22	18	0.000
Sternopygidae	0.27	0.23	0.23	0.27	0.47	180332.83	128.67	33	0.000

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Table S3. Continued. Base frequencies and percentage of GC (%GC) content in each major lineage and family across all filtered and unfiltered data sets of the full taxon set. Chi-Square homogeneity tests *p*-values in bold indicate significance at *P*<0.05.

	Base								
Taxon	A	C	G	T	GC	No. sites	χ^2	d.f.	P
QIRP 1 st + 2 nd quartiles									
All	0.27	0.23	0.23	0.27	0.46	188475.55	1067.11	138	0.000
Ingroup	0.27	0.23	0.23	0.27	0.46	190500.79	485.87	123	0.000
Apteronotidae	0.27	0.23	0.23	0.27	0.45	196278.47	34.36	42	0.793
Gymnotidae	0.27	0.23	0.23	0.27	0.46	189788.12	29.18	21	0.110
<i>Gymnotus</i>	0.27	0.23	0.23	0.27	0.46	181506.17	24.24	15	0.061
Hypopomidae	0.27	0.23	0.23	0.27	0.46	191061.00	11.61	9	0.236
Rhamphichthyidae	0.27	0.23	0.23	0.27	0.46	180189.33	96.74	6	0.000
Rhamphichthyoidea	0.27	0.23	0.23	0.27	0.46	186401.71	123.90	18	0.000
Sternopygidae	0.27	0.23	0.23	0.27	0.46	186144.92	83.58	33	0.000
Parsimony Informative 1 st quartile									
All	0.27	0.23	0.23	0.27	0.46	103557.38	886.66	138	0.000
Ingroup	0.27	0.23	0.23	0.27	0.46	104598.48	458.48	123	0.000
Apteronotidae	0.27	0.23	0.23	0.27	0.46	107707.60	38.81	42	0.612
Gymnotidae	0.27	0.23	0.23	0.27	0.46	104047.00	26.02	21	0.206
<i>Gymnotus</i>	0.27	0.23	0.23	0.27	0.46	99781.83	24.24	15	0.061
Hypopomidae	0.27	0.23	0.23	0.27	0.47	106003.50	6.47	9	0.693
Rhamphichthyidae	0.27	0.23	0.23	0.27	0.46	97185.00	126.36	6	0.000
Rhamphichthyoidea	0.27	0.23	0.23	0.27	0.47	102224.14	140.36	18	0.000
Sternopygidae	0.27	0.23	0.23	0.27	0.46	102464.75	92.80	33	0.000
Molecular Clock 1 st quartile									
All	0.26	0.24	0.24	0.26	0.48	93656.49	1190.38	138	0.000
Ingroup	0.26	0.24	0.24	0.26	0.48	94360.24	391.10	123	0.000
Apteronotidae	0.26	0.24	0.24	0.26	0.48	97340.33	51.54	42	0.148
Gymnotidae	0.26	0.24	0.24	0.26	0.48	94987.38	15.21	21	0.812
<i>Gymnotus</i>	0.26	0.24	0.24	0.26	0.48	91276.83	10.62	15	0.779
Hypopomidae	0.26	0.24	0.24	0.26	0.49	94919.75	5.17	9	0.819
Rhamphichthyidae	0.26	0.24	0.24	0.26	0.48	90789.00	60.85	6	0.000
Rhamphichthyoidea	0.26	0.24	0.24	0.26	0.48	93149.43	72.75	18	0.000
Sternopygidae	0.26	0.24	0.24	0.26	0.48	90923.33	94.67	33	0.000
Phylogenetic Informativeness 1 st quartile									
All	0.27	0.23	0.23	0.27	0.46	101813.57	915.28	138	0.000
Ingroup	0.27	0.23	0.23	0.27	0.46	102906.74	484.39	123	0.000
Apteronotidae	0.27	0.23	0.23	0.27	0.46	106130.20	39.28	42	0.591
Gymnotidae	0.27	0.23	0.23	0.27	0.46	102184.00	28.57	21	0.125
<i>Gymnotus</i>	0.27	0.23	0.23	0.27	0.46	98127.33	27.25	15	0.027
Hypopomidae	0.27	0.23	0.23	0.27	0.47	104272.50	10.02	9	0.349
Rhamphichthyidae	0.27	0.23	0.23	0.27	0.46	96100.00	128.22	6	0.000
Rhamphichthyoidea	0.27	0.23	0.23	0.27	0.47	100770.00	146.29	18	0.000
Sternopygidae	0.27	0.23	0.23	0.27	0.46	100605.67	97.51	33	0.000

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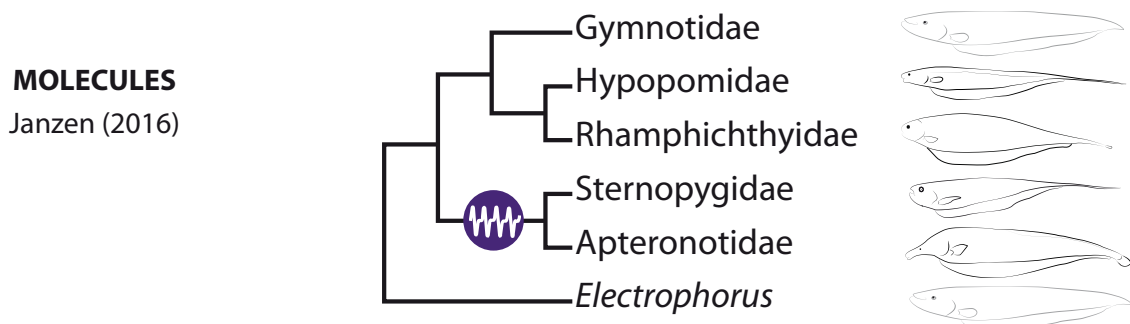
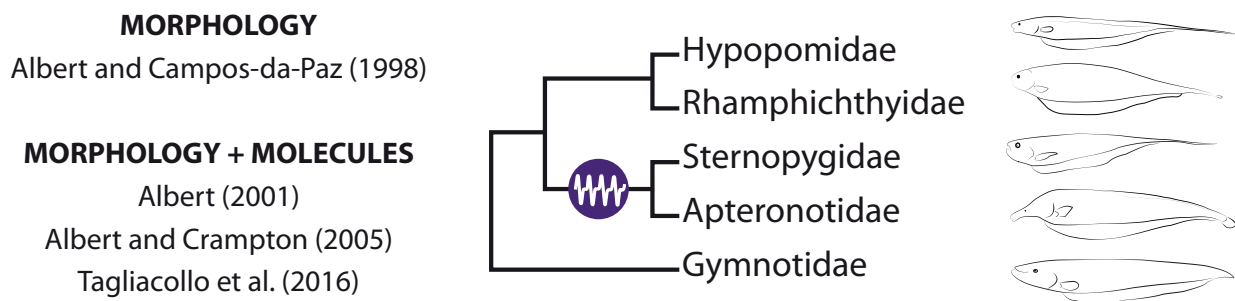
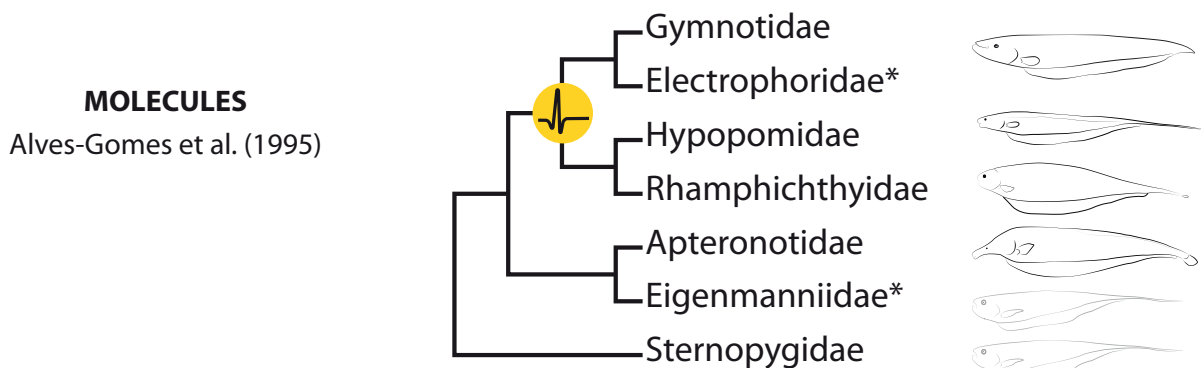
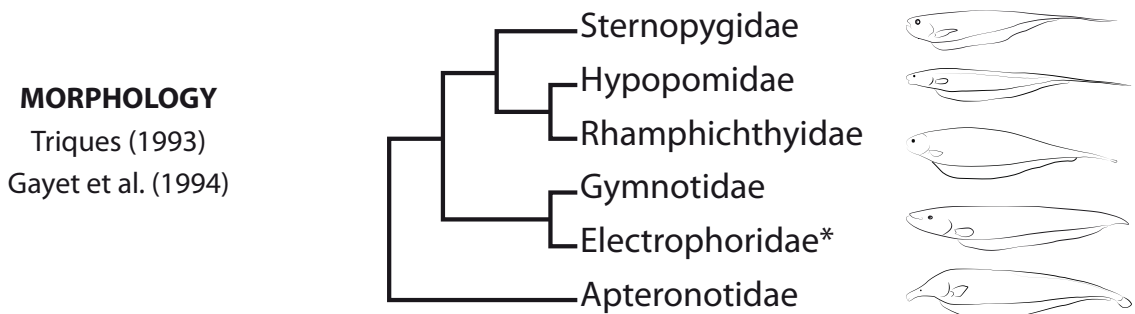
Table S3. Continued. Base frequencies and percentage of GC (%GC) content in each major lineage and family across all filtered and unfiltered data sets of the full taxon set. Chi-Square homogeneity tests *p*-values in bold indicate significance at *P*<0.05.

Significance at $P < 0.05$.									
	Base								
Taxon	A	C	G	T	GC	No. sites	χ^2	d.f.	<i>P</i>
QIRP 1st quartile									
All	0.27	0.22	0.22	0.28	0.45	106752.43	697.40	138	0.000
Ingroup	0.27	0.22	0.22	0.28	0.45	108147.38	359.26	123	0.000
Apterodontidae	0.27	0.22	0.22	0.28	0.44	110946.27	37.69	42	0.660
Gymnotidae	0.27	0.22	0.22	0.28	0.44	108107.75	18.14	21	0.640
<i>Gymnotus</i>	0.27	0.22	0.22	0.28	0.44	104018.50	13.29	15	0.580
Hypopomidae	0.27	0.23	0.23	0.28	0.45	108818.50	8.46	9	0.489
Rhamphichthyidae	0.27	0.23	0.22	0.28	0.45	102130.00	68.28	6	0.000
Rhamphichthyoidea	0.27	0.23	0.22	0.28	0.45	105952.00	85.36	18	0.000
Sternopygidae	0.27	0.23	0.22	0.28	0.45	105955.83	63.41	33	0.001

Table S4. Results from the Linear Mixed Effects Models used to test for significant differences in GC composition among filtered and non-filtered data sets.

Fixed effects	Estimate	Std. Error	d.f.	<i>t</i> value	<i>P</i>
Initial linear mixed effects model (BIC = -3501.53)					
Molecular Clock 1 st quartile	5.15E-03	4.81E-04	354	10.70	<2E-16
Molecular Clock 1 st + 2 nd quartiles	3.81E-03	4.81E-04	354	7.92	2.22E-15
Parsimony Informative 1 st quartile	-1.73E-02	4.59E-04	354	-37.73	<2E-16
Parsimony Informative 1 st + 2 nd quartiles	-1.18E-02	4.59E-04	354	-25.76	<2E-16
Phylogenetic Informativeness 1 st quartile	-1.82E-02	4.59E-04	354	-39.65	<2E-16
Phylogenetic Informativeness 1 st + 2 nd quartile	-1.12E-02	4.59E-04	354	-24.41	<2E-16
QIRP 1 st quartiles	-3.00E-02	4.59E-04	354	-65.45	<2E-16
QIRP 1 st and 2 nd quartile	-1.99E-02	4.59E-04	354	-43.48	<2E-16
Post-hoc comparisons: Tukey Contrasts					
	Estimate	Std. Error		<i>z</i> value	<i>P</i>
Molecular Clock 1 st quartile - Non Filtered	5.15E-03	4.81E-04		10.70	<0.001
Molecular Clock 1 st + 2 nd quartiles - Non Filtered	3.81E-03	4.81E-04		7.92	<0.001
Parsimony Informative 1 st quartile - Non Filtered	-1.73E-02	4.59E-04		-37.73	<0.001
Parsimony Informative 1 st + 2 nd quartiles - Non Filtered	-1.18E-02	4.59E-04		-25.76	<0.001
Phylogenetic Informativeness 1 st quartile - Non Filtered	-1.82E-02	4.59E-04		-39.65	<0.001
Phylogenetic Informativeness 1 st + 2 nd quartiles - Non Filtered	-1.12E-02	4.59E-04		-24.41	<0.001
QIRP 1 st quartile - Non Filtered	-3.00E-02	4.59E-04		-65.45	<0.001
QIRP 1 st + 2 nd quartiles - Non Filtered	-1.99E-02	4.59E-04		-43.48	<0.001
Molecular Clock 1 st quartile - Molecular Clock 1 st + 2 nd quartiles	-1.33E-03	4.59E-04		-2.91	0.086
Parsimony Informative 1 st quartile - Parsimony Informative 1 st + 2 nd quartiles	5.49E-03	4.59E-04		11.98	<0.001
Phylogenetic Inf. 1 st quartile - Phylogenetic Inf. 1 st + 2 nd quartiles - Non Filtered	6.99E-03	4.59E-04		15.24	<0.001
QIRP 1 st quartile - QIRP 1 st + 2 nd quartiles	1.01E-02	4.59E-04		21.97	<0.001

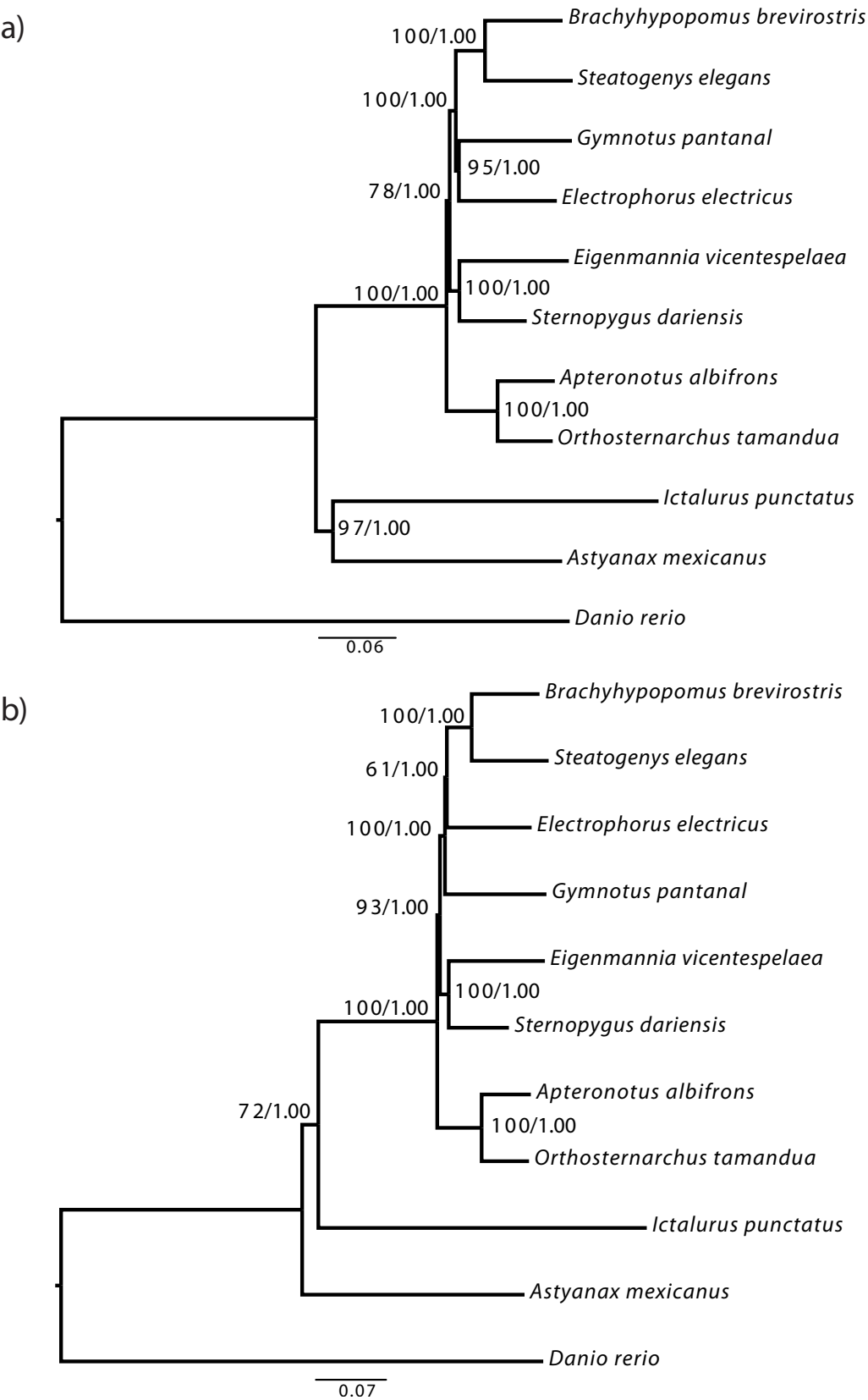
Suppl. Fig. S1. Summary of previously recovered hypotheses of Gymnotiformes relationships. Hypotheses that recovered monophyletic Sinusoidea or Pulseoidea are indicated with symbols of electric organ discharges in purple and gold, respectively. Asterisks indicate families that are not accepted in the current taxonomy. Fish line drawings by Maxwell J. Bernt.



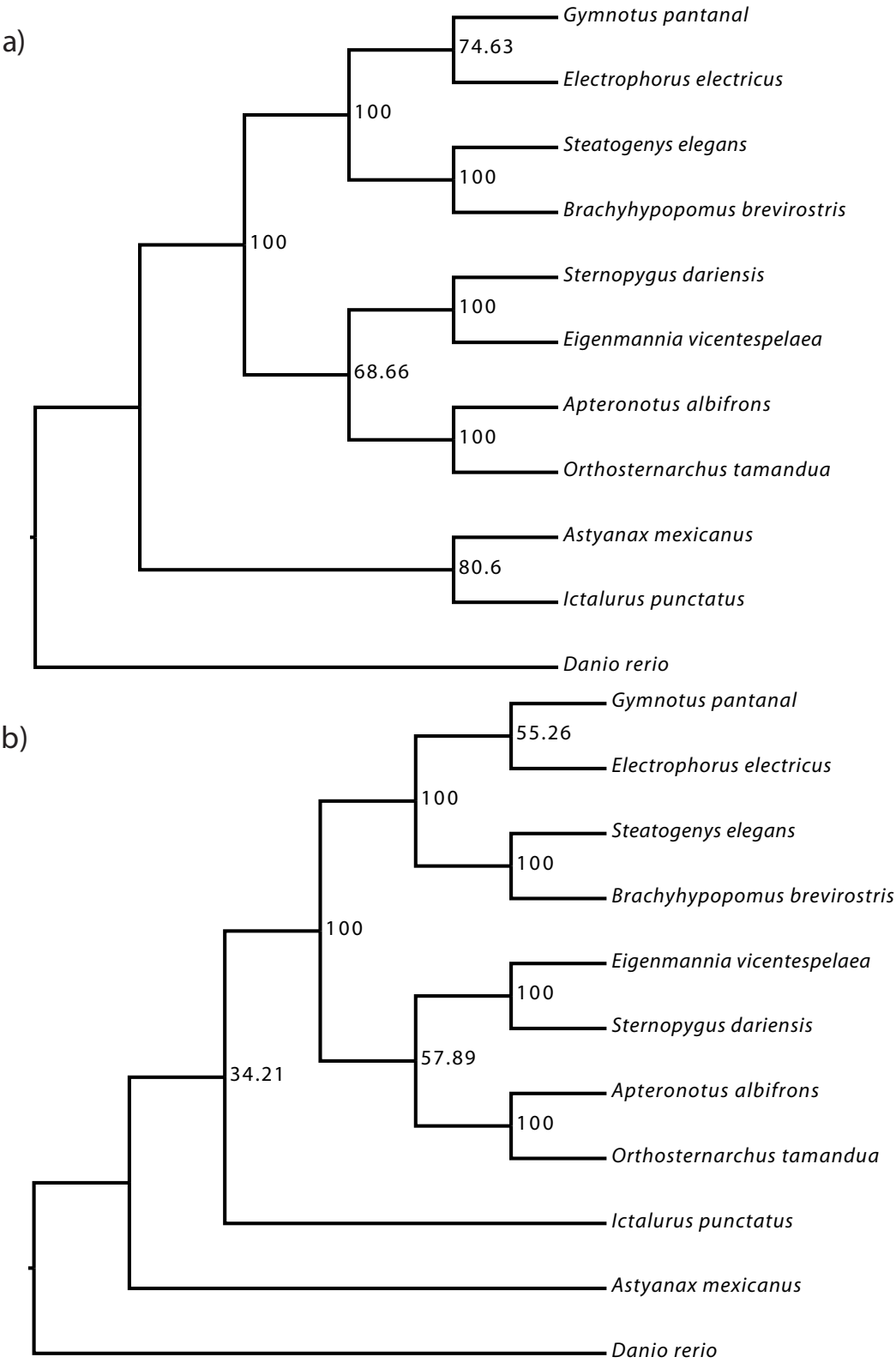
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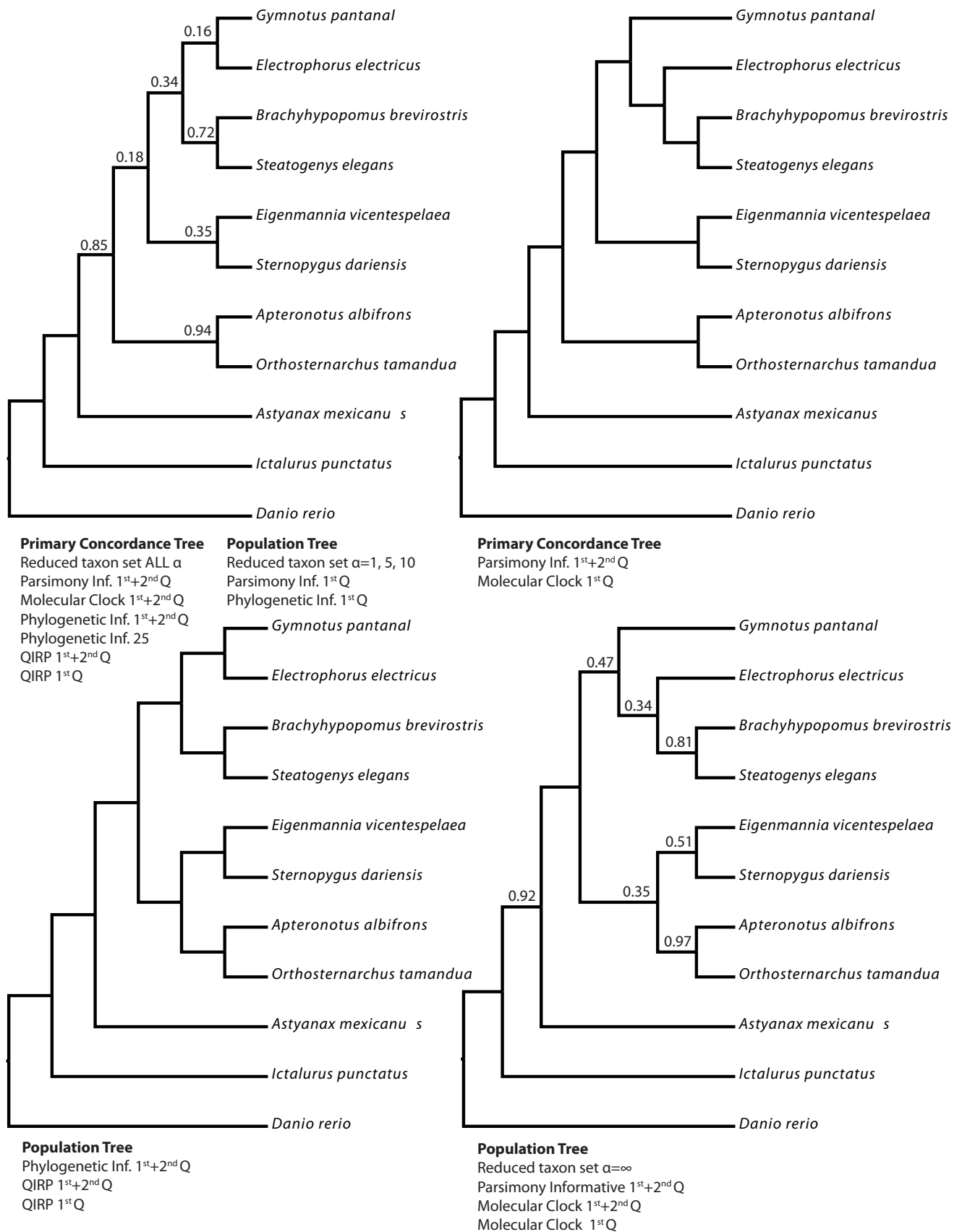
Suppl. Fig. S2. Maximum Likelihood tree for reduced taxon sets with 25% missing data (a) and no missing data (b).



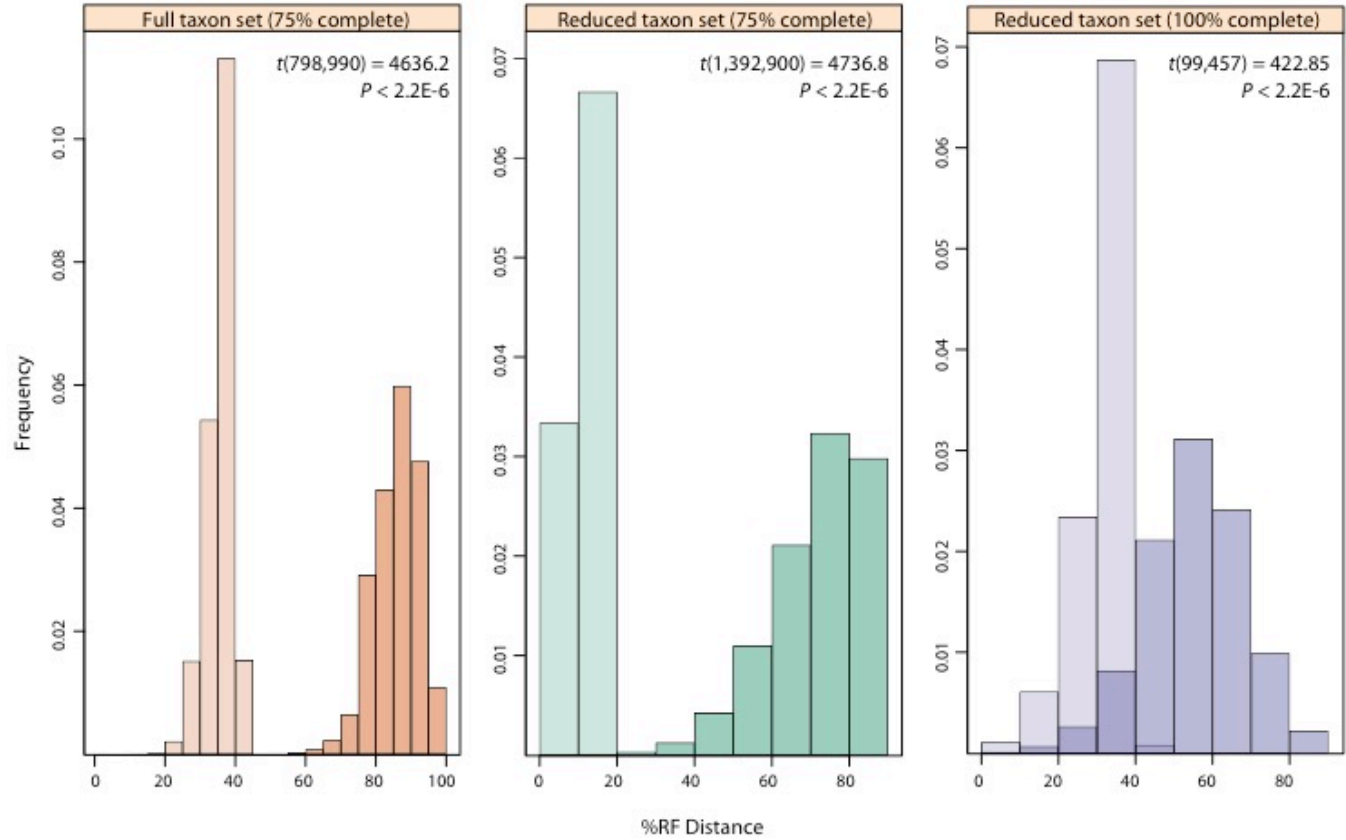
Suppl. Fig. S3. ASTRAL-II species tree for reduced taxon sets with 25% missing data (a) and no missing data (b).



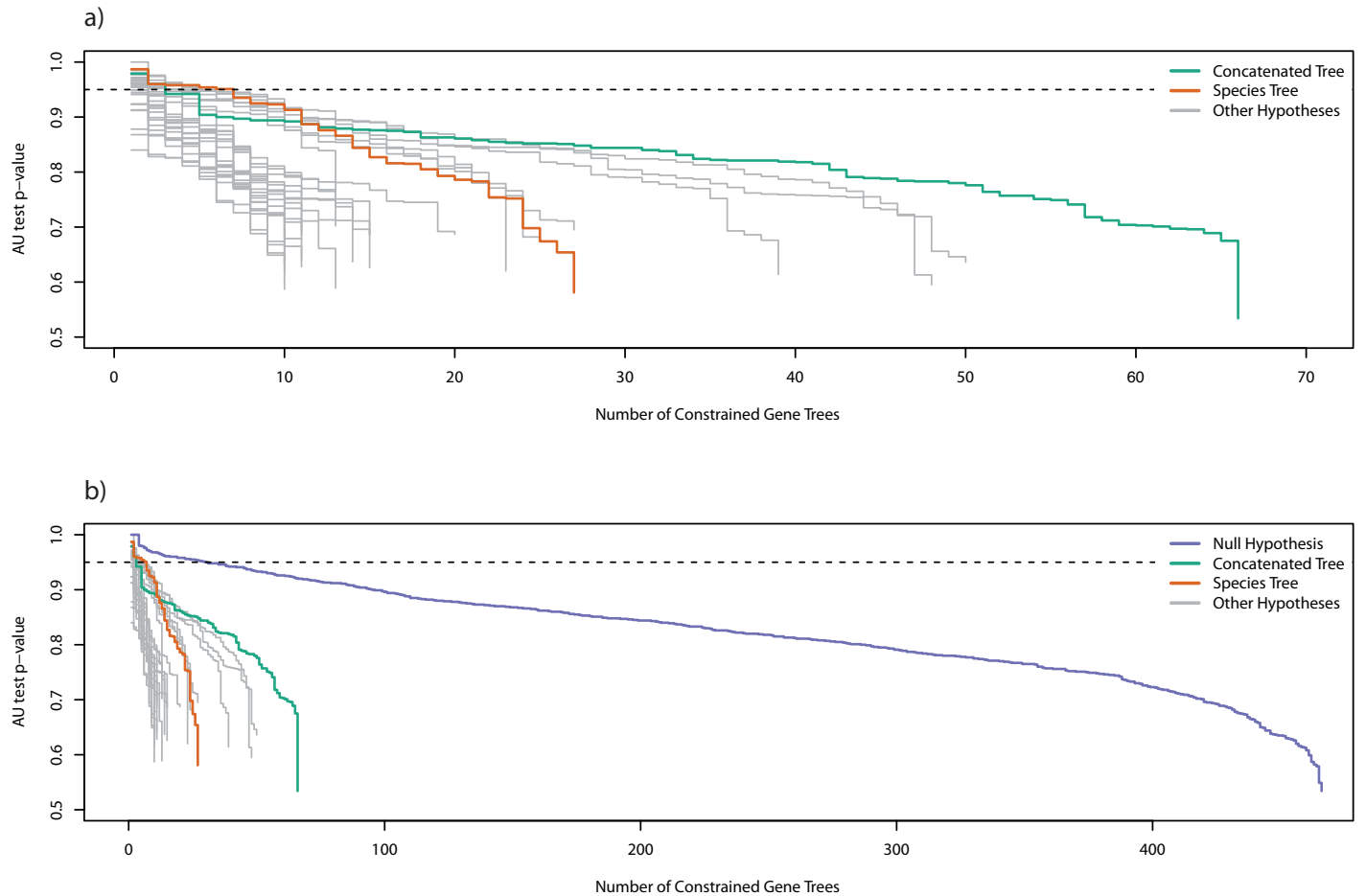
Suppl. Fig. S4. Primary concordance and population trees inferred from Bayesian concordance analysis in BUCKy for the non-filtered and filtered data sets. Concordance values are indicated in the trees inferred for the non-filtered data sets using $\alpha=\infty$.



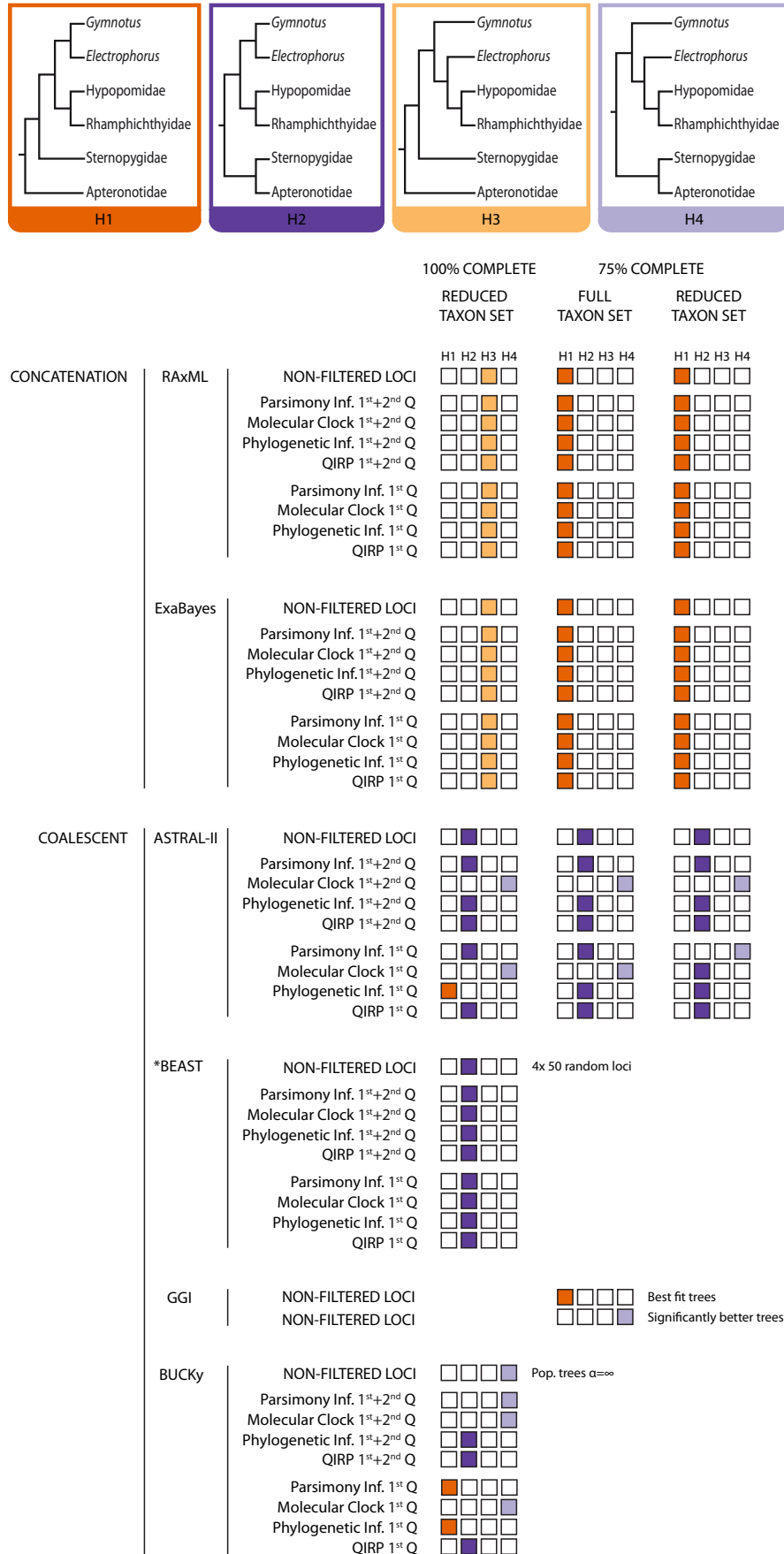
Suppl. Fig. S5. Histograms showing the distribution of gene trees expected simulated under thecoalescent model for the inferred species trees (light colors), and of the observed gene trees (dark colors) in each of the analyzed data sets for the full taxon set. Values for the Student’s *t*-test and their associated *P*-values are shown in for each plot.



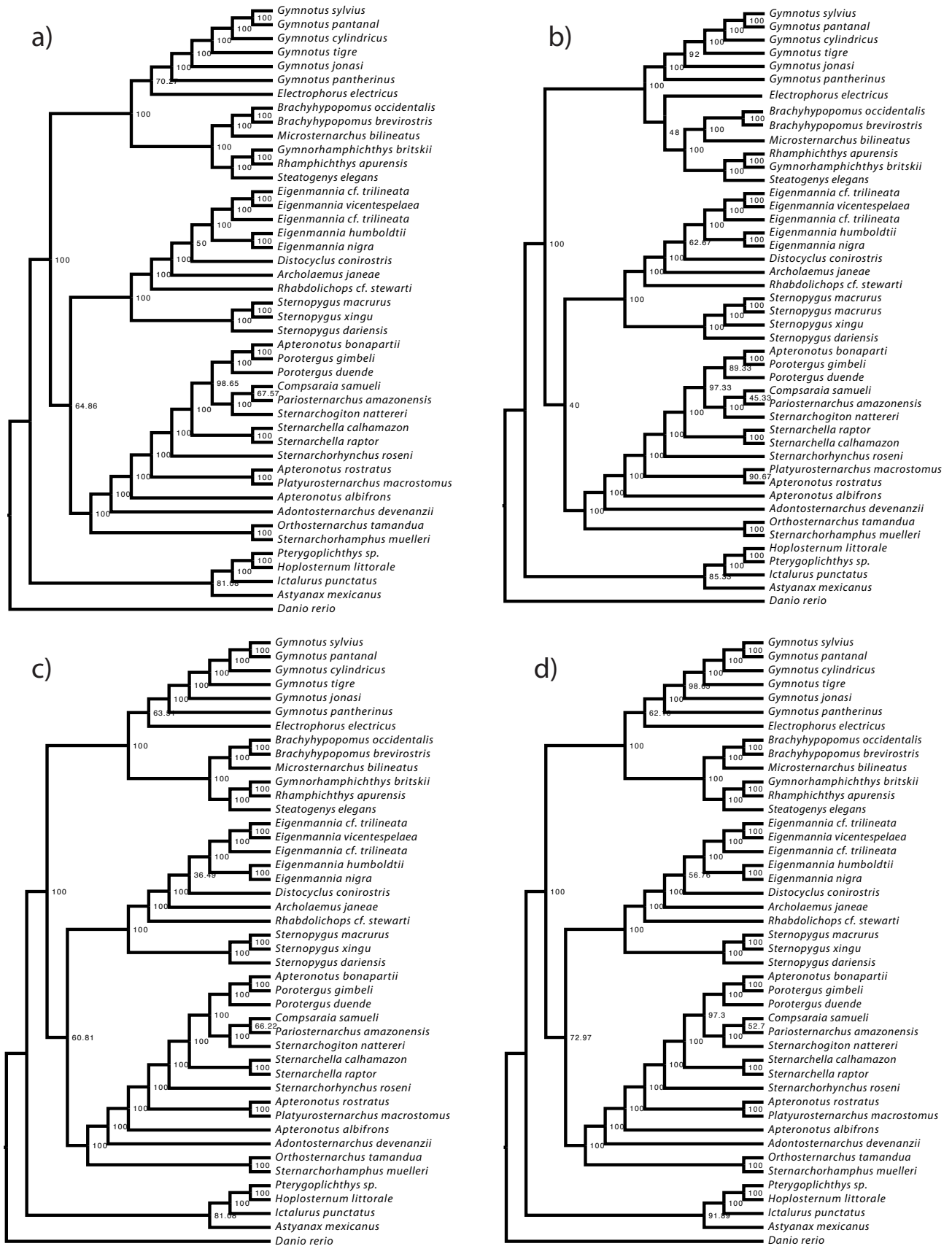
Suppl. Fig. S6. Results plot from the Gene Genealogy Interrogation (GGI) analysis testing 105 alternative hypotheses for the relationships among families of Gymnotiformes (a), and 106 hypotheses (i.e., 105 hypotheses + the null hypothesis of a polytomy at the base of the tree) (b). The x axis represents the cumulative number of loci supporting each hypothesis. The green line indicates the topology supported by the concatenated phylogenomic analyses, the orange line represents the topology of the species tree obtained in the coalescent analyses, and the blue line is the null hypothesis. The y axis represents associated P -values from the approximately unbiased test (AU) for each topology. Only values above the dashed line are hypotheses significantly better than the alternatives.



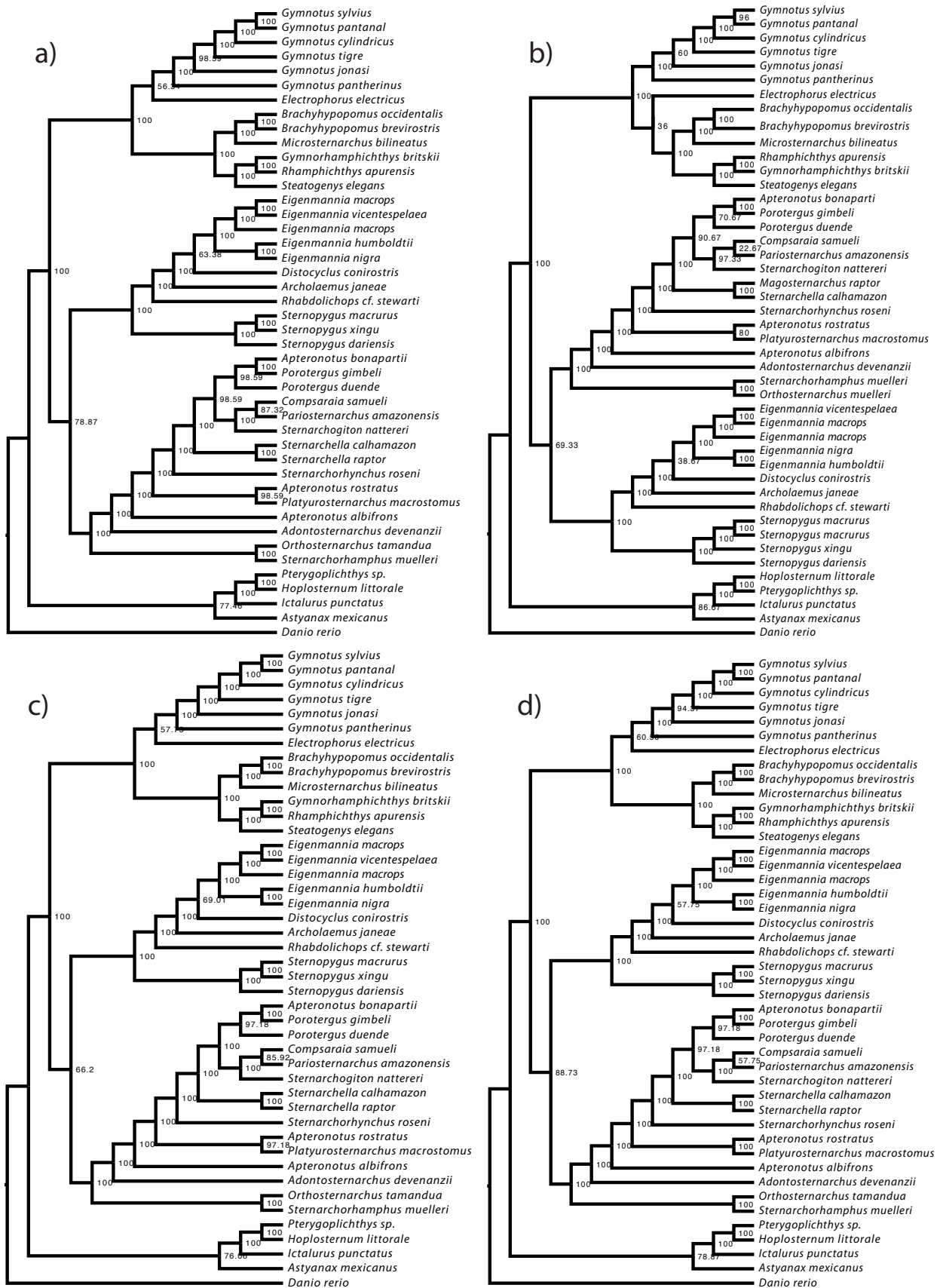
Suppl. Fig. S7. Summary of phylogenetic hypotheses recovered across all analyses and data sets.



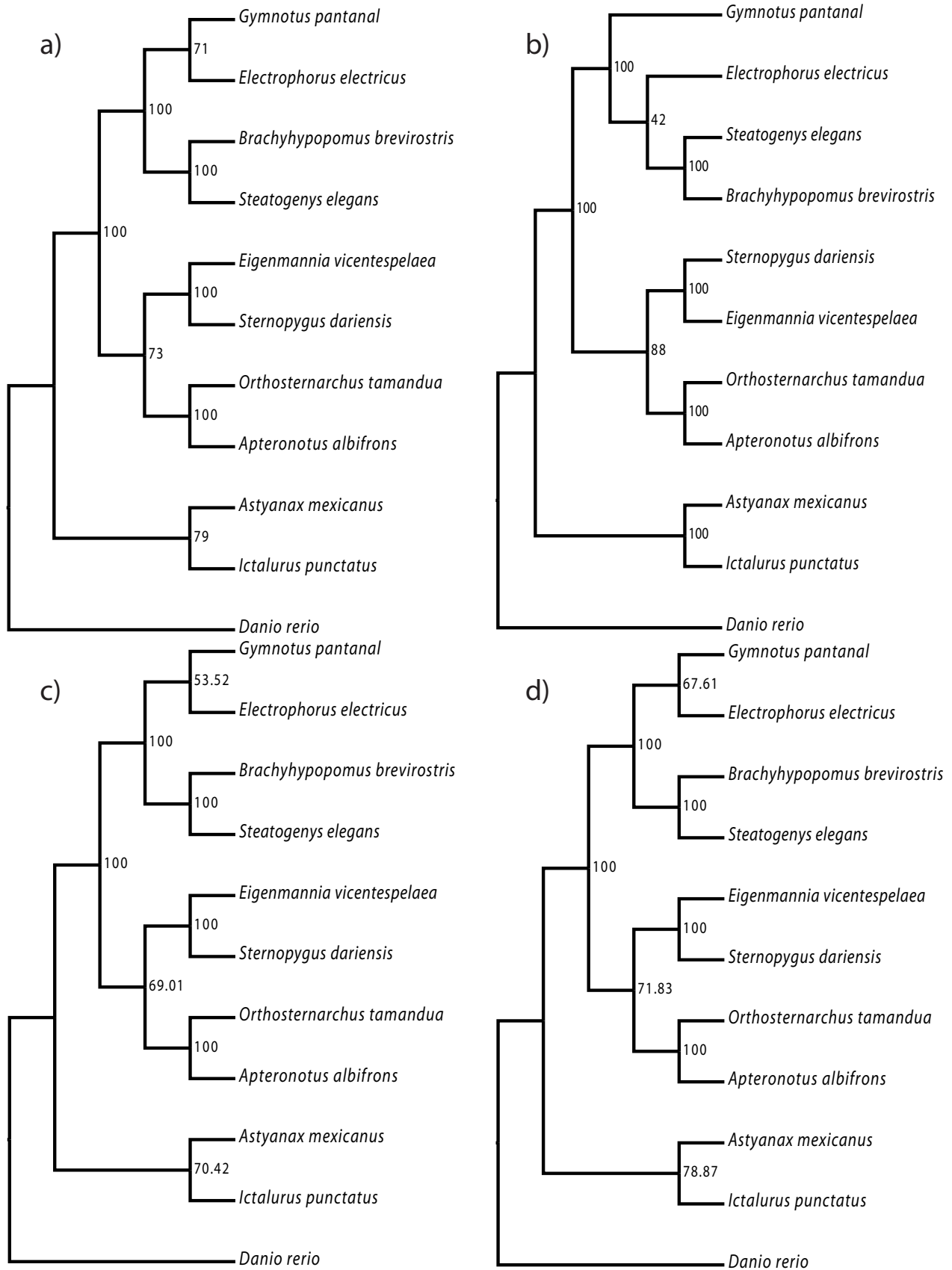
Suppl. Fig. S8. ASTRAL-II species trees inferred for the full taxon set data using the 1st and 2nd quartiles of loci filtered according to the number of parsimony informative sites (a), clock-likeness (b), phylogenetic informativeness (c), and QIRP (d).



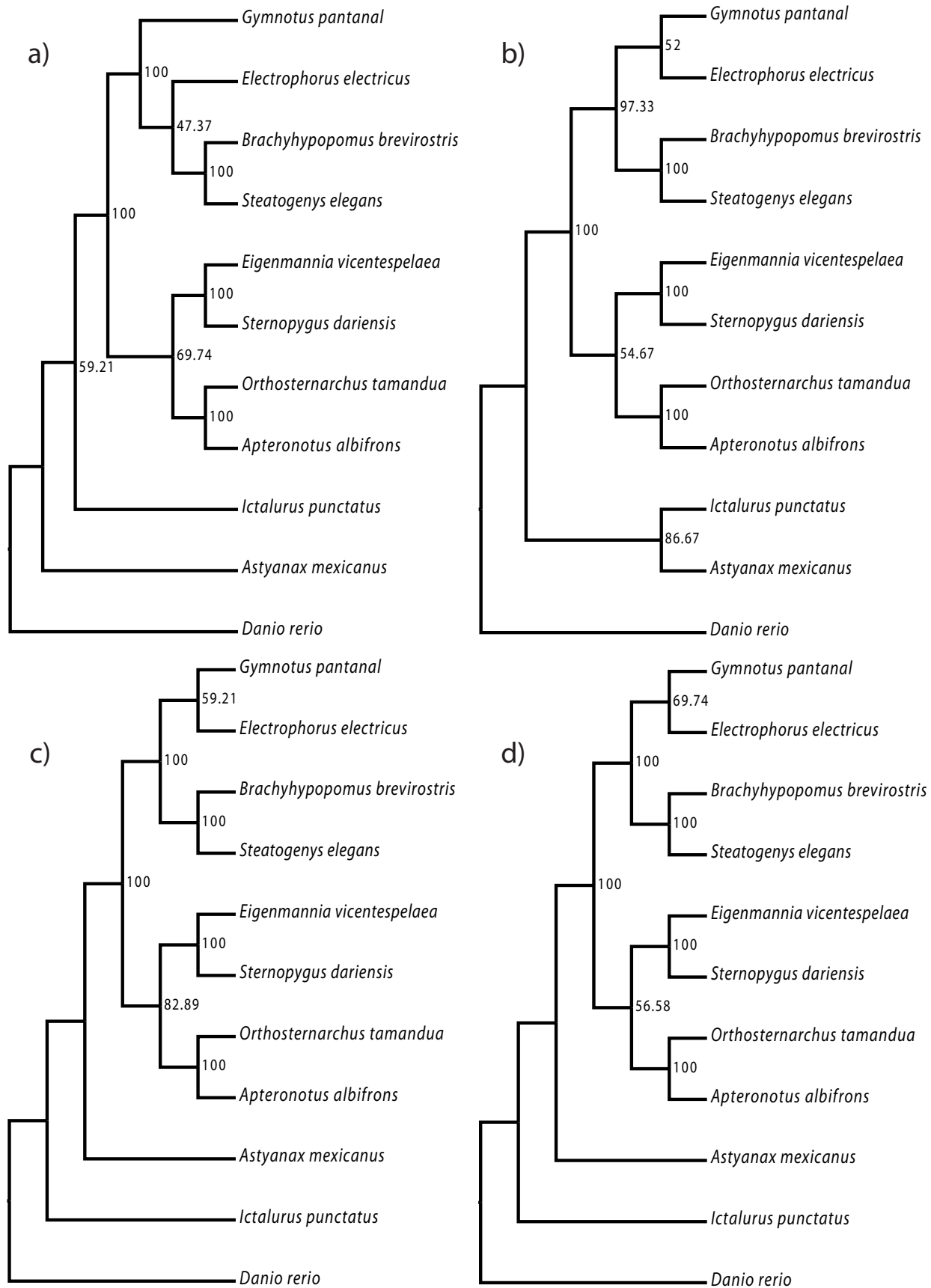
Suppl. Fig. S9. ASTRAL-II species trees inferred for the full taxon set data using the 1st quartile of loci filtered according to the number of parsimony informative sites (a), clock-likeness (b), phylogenetic informativeness (c), and QIRP (d).



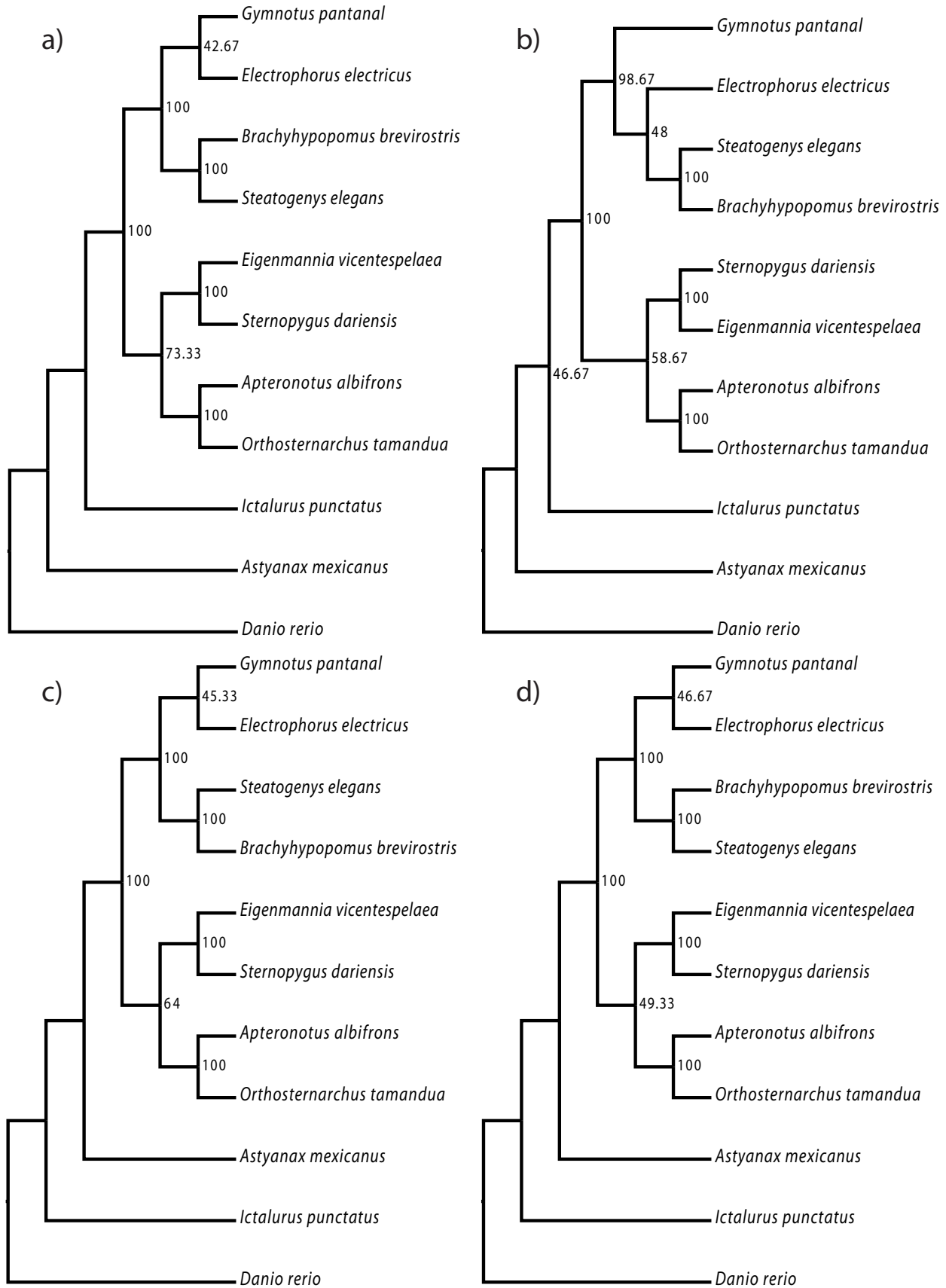
Suppl. Fig. S10. ASTRAL-II species trees inferred for the reduced taxon set with 25% of missing data using the 1st and 2nd quartiles of loci filtered according to the number of parsimony informative sites (a), clock-likeness (b), phylogenetic informativeness (c), and QIRP (d).



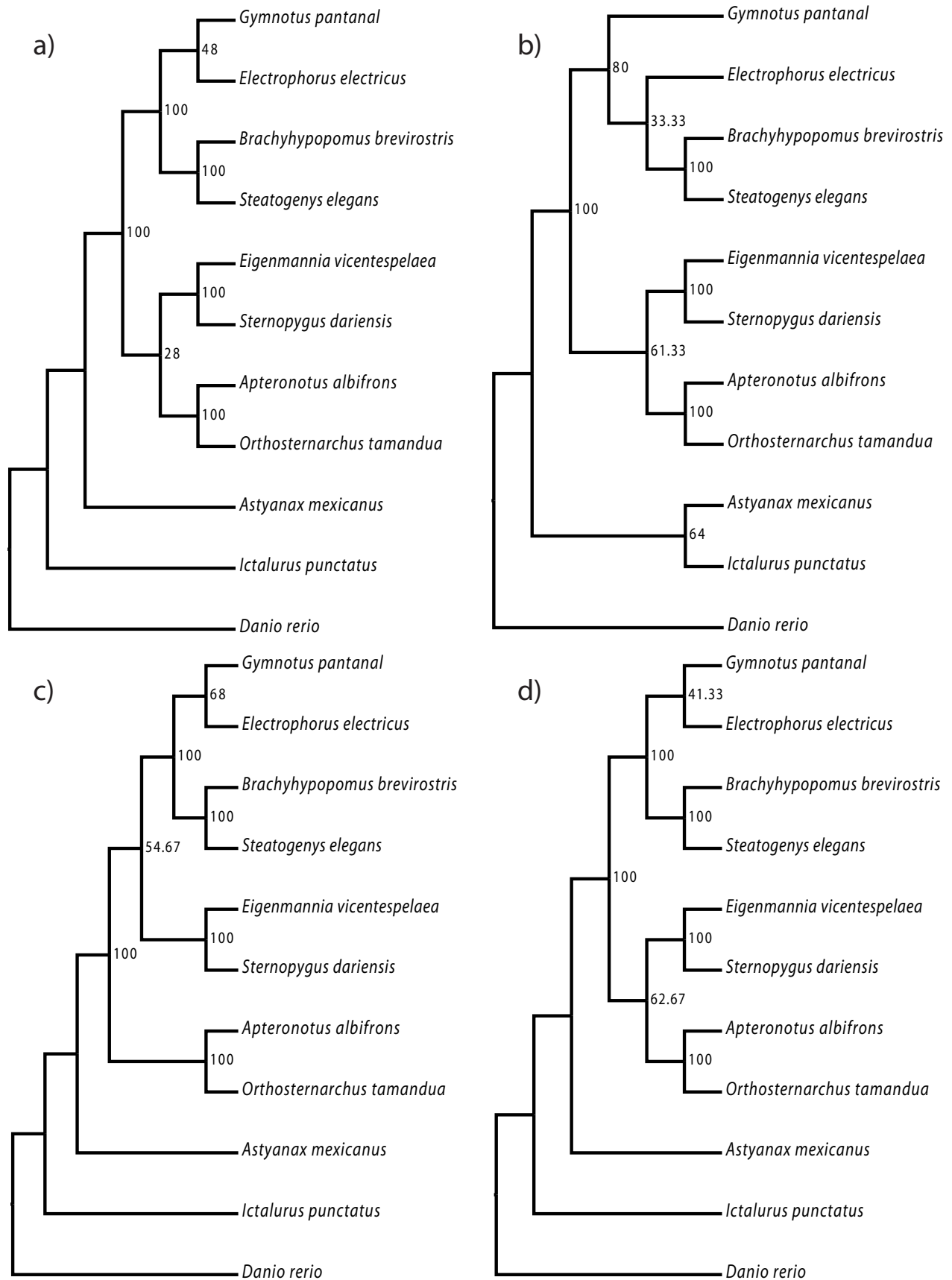
Suppl. Fig. S11. ASTRAL-II species trees inferred for the reduced taxon set with 25% of missing data using the 1st quartile of loci filtered according to the number of parsimony informative sites (a), clock-likeness (b), phylogenetic informativeness (c), and QIRP (d).



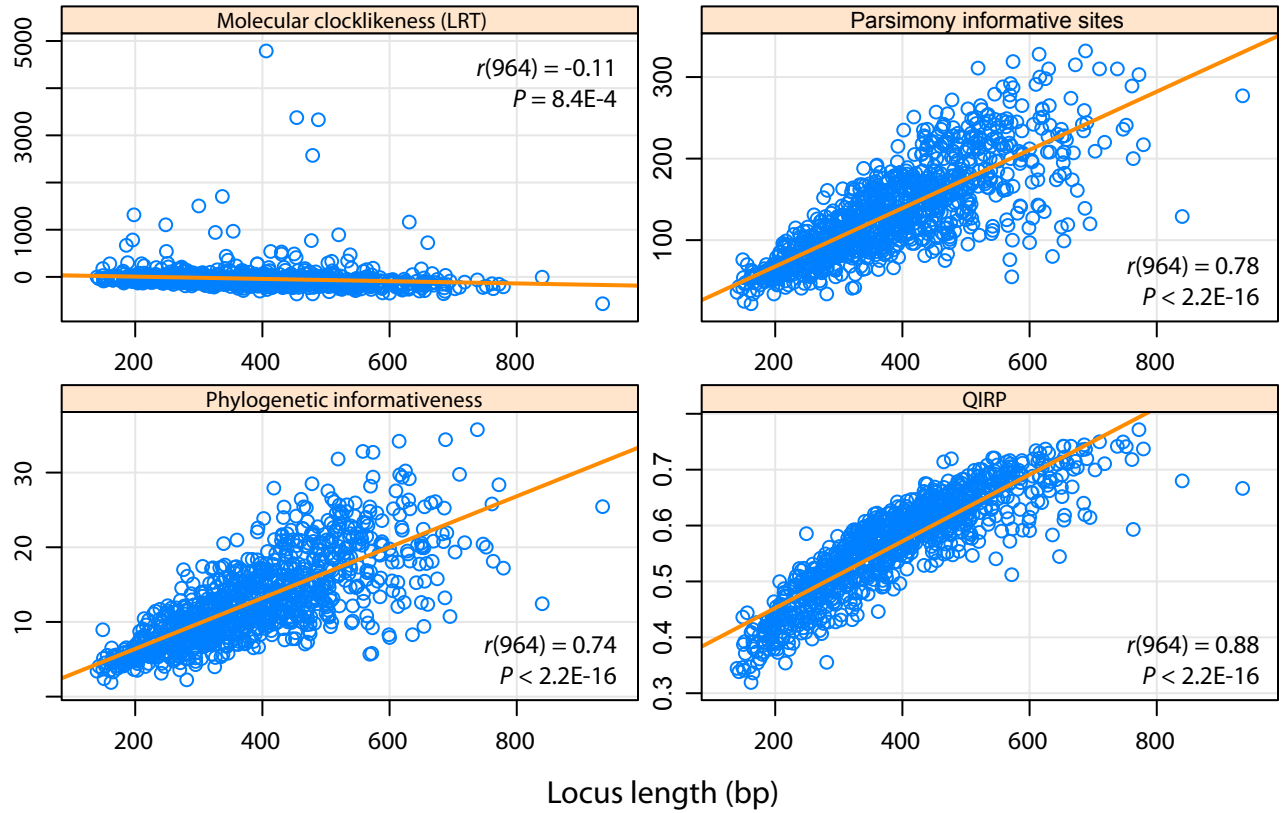
Suppl. Fig. S12. ASTRAL-II species trees inferred for the reduced taxon set and no missing data using the 1st and 2nd quartiles of loci filtered according to the number of parsimony informative sites (a), clock-likeness (b), phylogenetic informativeness (c), and QIRP (d).



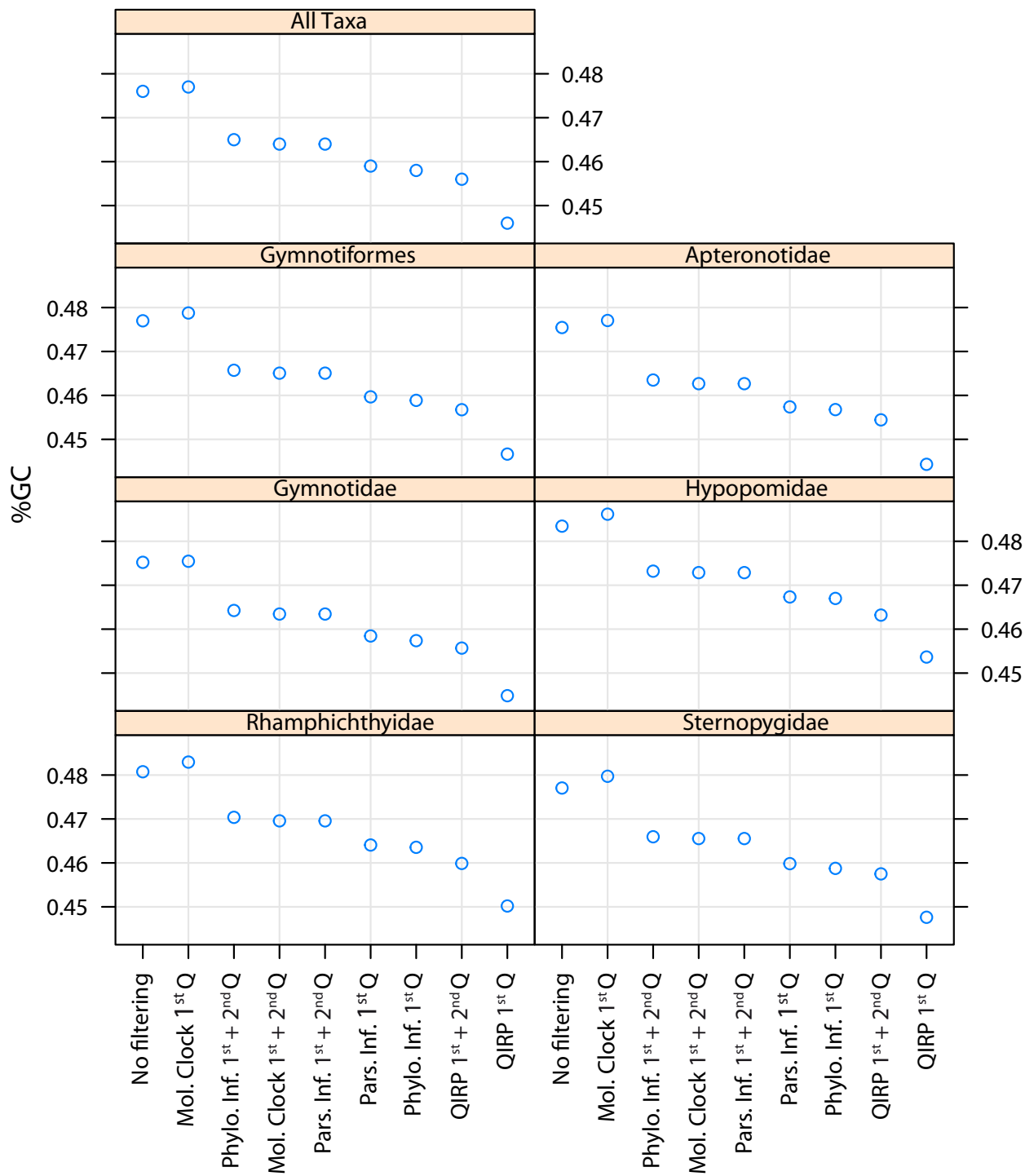
Suppl. Fig. S13. ASTRAL-II species trees inferred for the reduced taxon set and no missing data using the 1st quartile of loci filtered according to the number of parsimony informative sites (a), clock-likeness (b), phylogenetic informativeness (c), and QIRP (d).



Suppl. Fig. S14. Scatterplots showing the relationship between locus length and each of the criteria utilized to filter the complete taxon set. Values of Pearson's r correlations and associated P -values are indicated for each plot. Regression lines are shown in orange.



Suppl. Fig. S15. Graphical representation of the reduction in GC content (%GC) for each family of Gymnotiformes following locus filtering of the complete taxon set.



Suppl. Fig. S16. GGI species trees estimated in ASTRAL-II for the best fit trees (a), and for only the best trees that are significantly better than the second-ranked tree (b).

